

Machine Learning - Deep Learning fundamentals (69152) CNNs

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Reminders

- **Reading Lab (Lab 4)**
 - Pick a paper (tomorrow)
 - read it
 - prepare a presentation for your lab session

Today

- Open problems with deep learning?
- More well-known architectures

Compiled “open problems” from moodle answers

Ethics and Fairness : Is it possible to avoid bias? “Responsible AI”? External seminar

Explainability : finer-grain, probabilities, ...

Generalization - Adaptability : Incremental, “foundation-models”

Efficiency (time-memory) : efficient architectures, ENERGY?

Efficiency (data-requirements) : less supervision

Multi-modal

Meta-learning

Robustness

Open or interesting problems related to deep learning?

perform more complex tasks?

More architectures ...

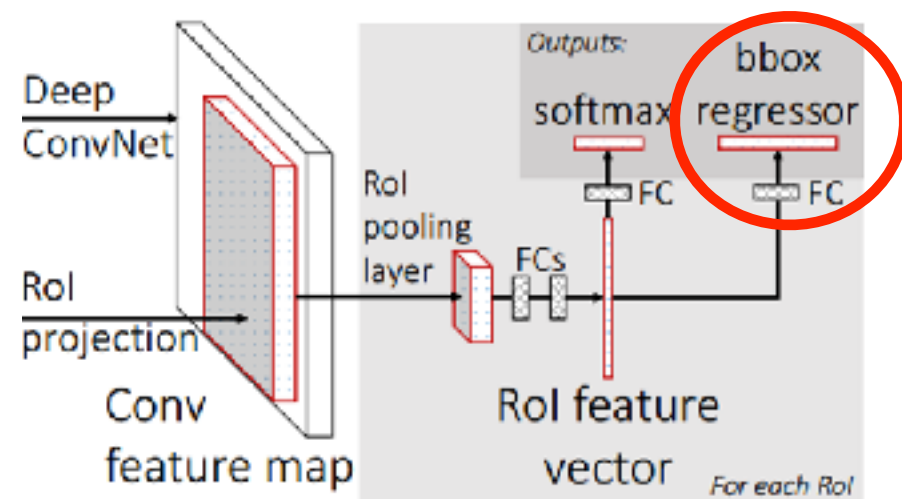
- Not only classification —> detection?

Deep Learning & Regression

- **Detection**

predict normalized b-box coordinates

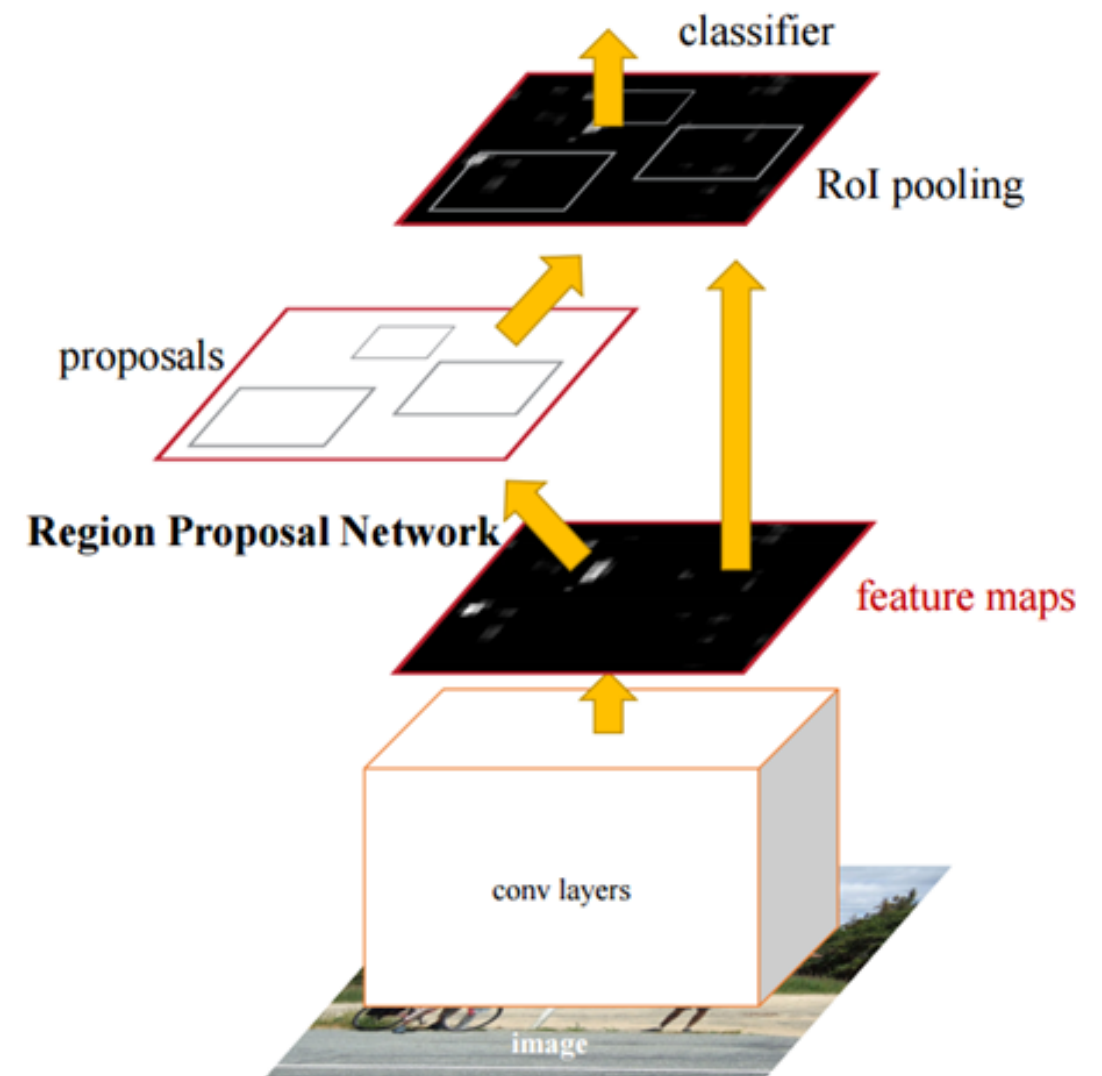
$$p = (x_1, y_1, x_2, y_2)$$



Fast R-CNN. Ross Girshick. 2015

Different versions of Region-CNN:

- R-CNN - 2013
- Fast R-CNN - 2015
- Faster R-CNN - 2016



CLASSIFICATION + DETECTION

Analyze feature maps (activations) to learn where the objects are

Deep Learning & Regression

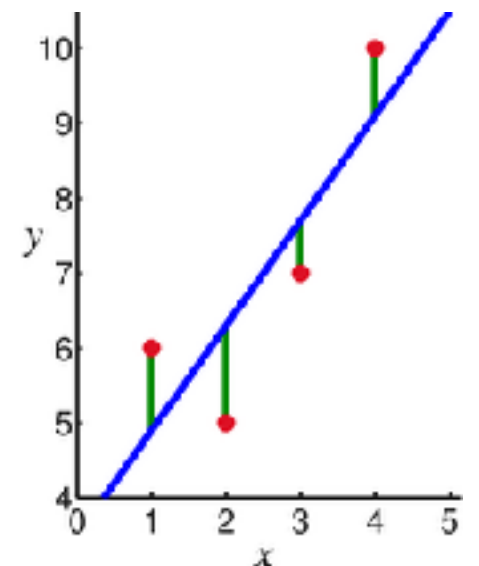
- Linear regression: relationship between a scalar (y) and one or more explanatory variables (x)

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad i=1, \dots, n$$

“map” to a continuous output (regression)

VS

“map” to a discrete output (classification)



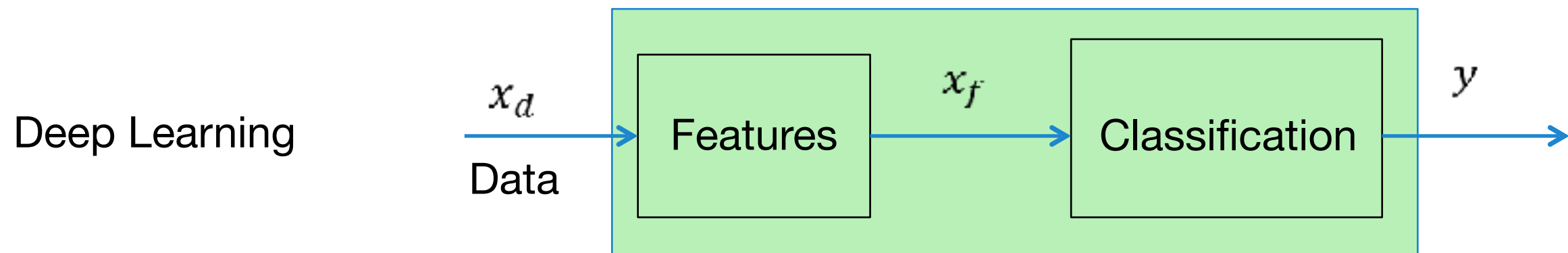
Wikipedia:
File:Linear least squares example2.png

Deep Learning & Regression

- Regression of **Pose** - *PoseNet*
 - Train the network to output 3D position ($\mathbf{x}$) and orientation ($\mathbf{q}$) (7 dims). Loss function:

$$loss(I) = \|\hat{\mathbf{x}} - \mathbf{x}\|_2 + \beta \left\| \hat{\mathbf{q}} - \frac{\mathbf{q}}{\|\mathbf{q}\|} \right\|_2$$

- Modify GoogLeNet:



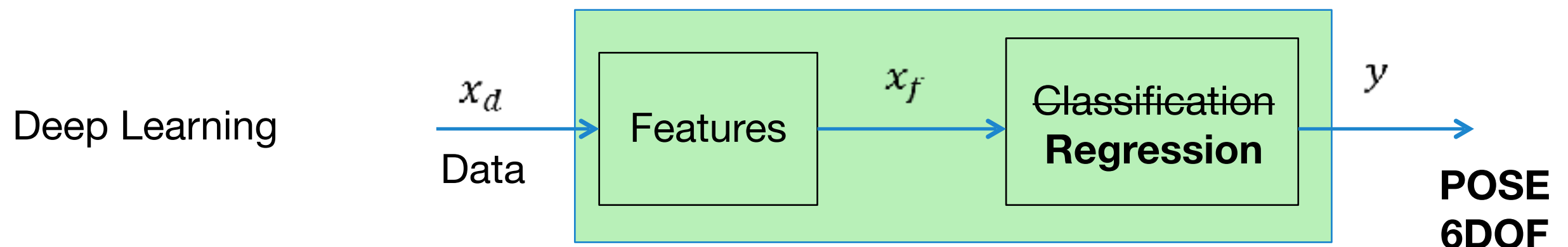
Deep Learning & Regression

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- Modify GoogLeNet. Key change: replace softmax classifiers with affine regressors (each final fully connected layer now outputs a pose vector of 7-dims).

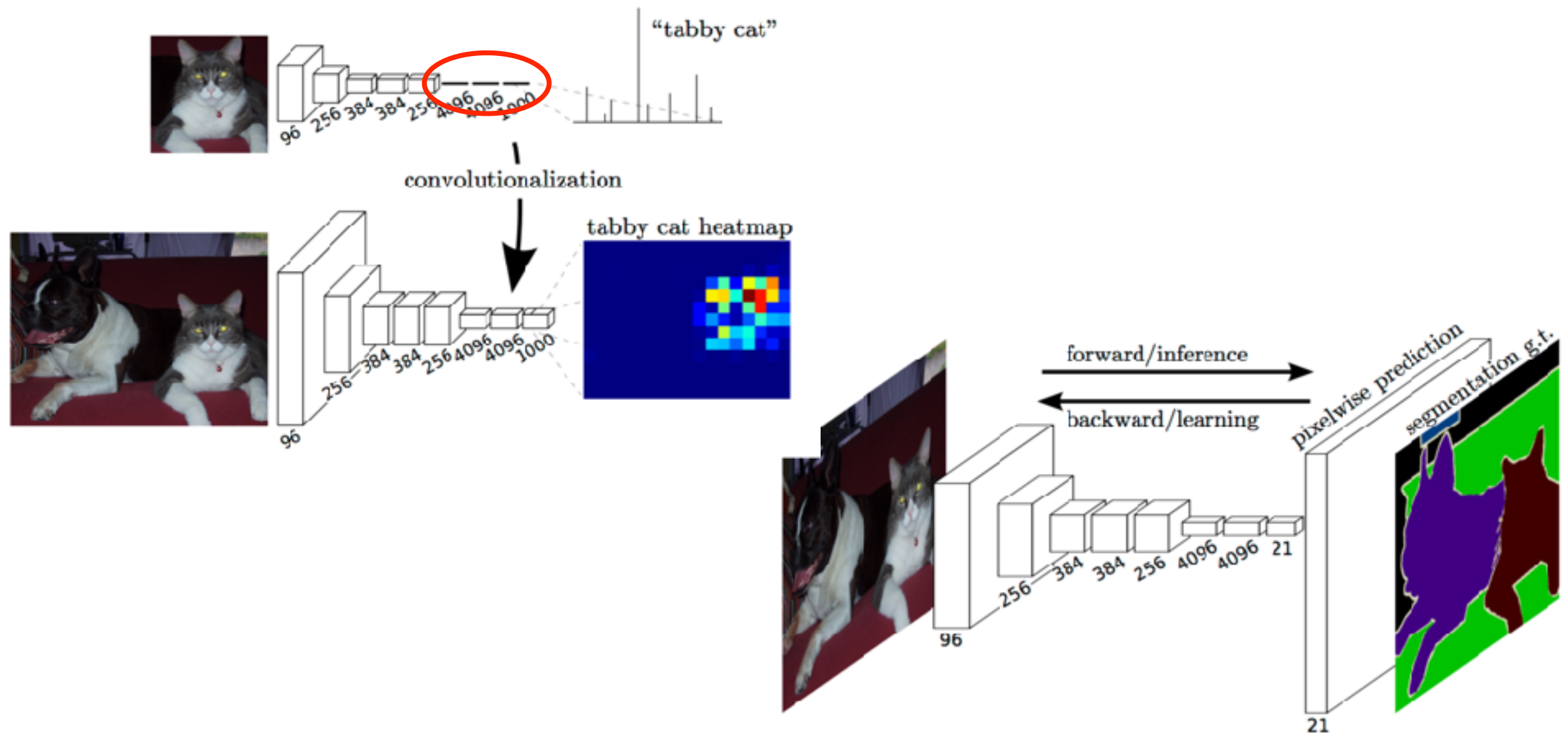


More architectures ...

More accurate image understanding?

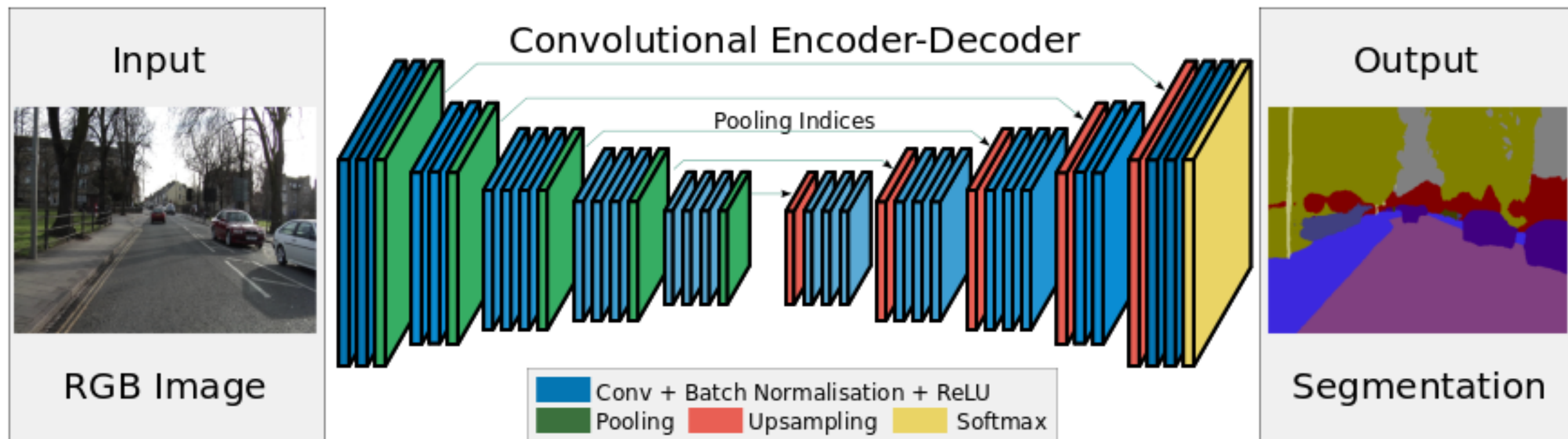
Deep Learning & pixel classification

- Dense labeling/**Segmentation** - Fully Convolutional Net (FCN)



Deep Learning & pixel classification

- Dense labeling/**Segmentation** - Encoder-Decoder



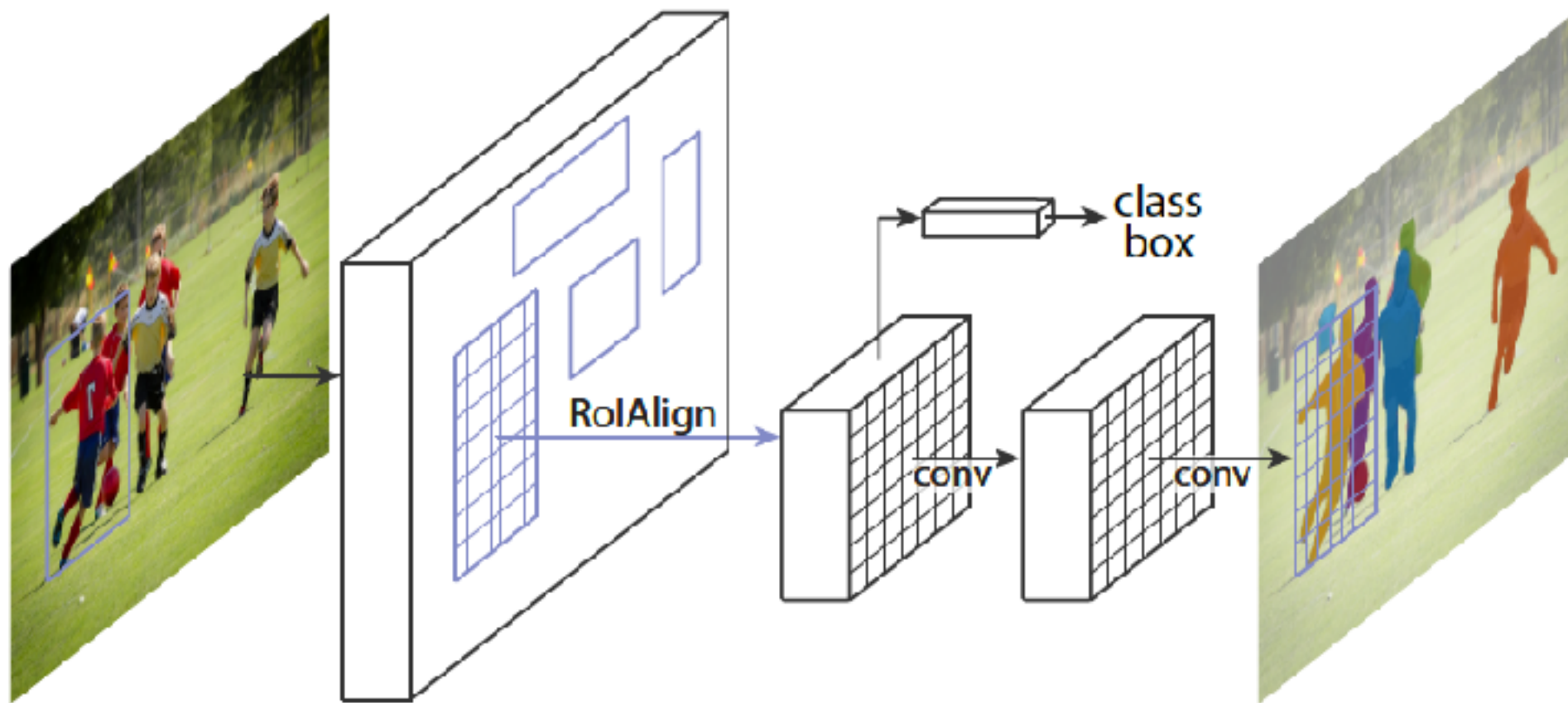
<http://mi.eng.cam.ac.uk/projects/segnet/>

Deep Learning & pixel classification

- **Detection + Instance Segmentation**

Mask R-CNN - 2017

(R-CNN + Semantic Segmentation)

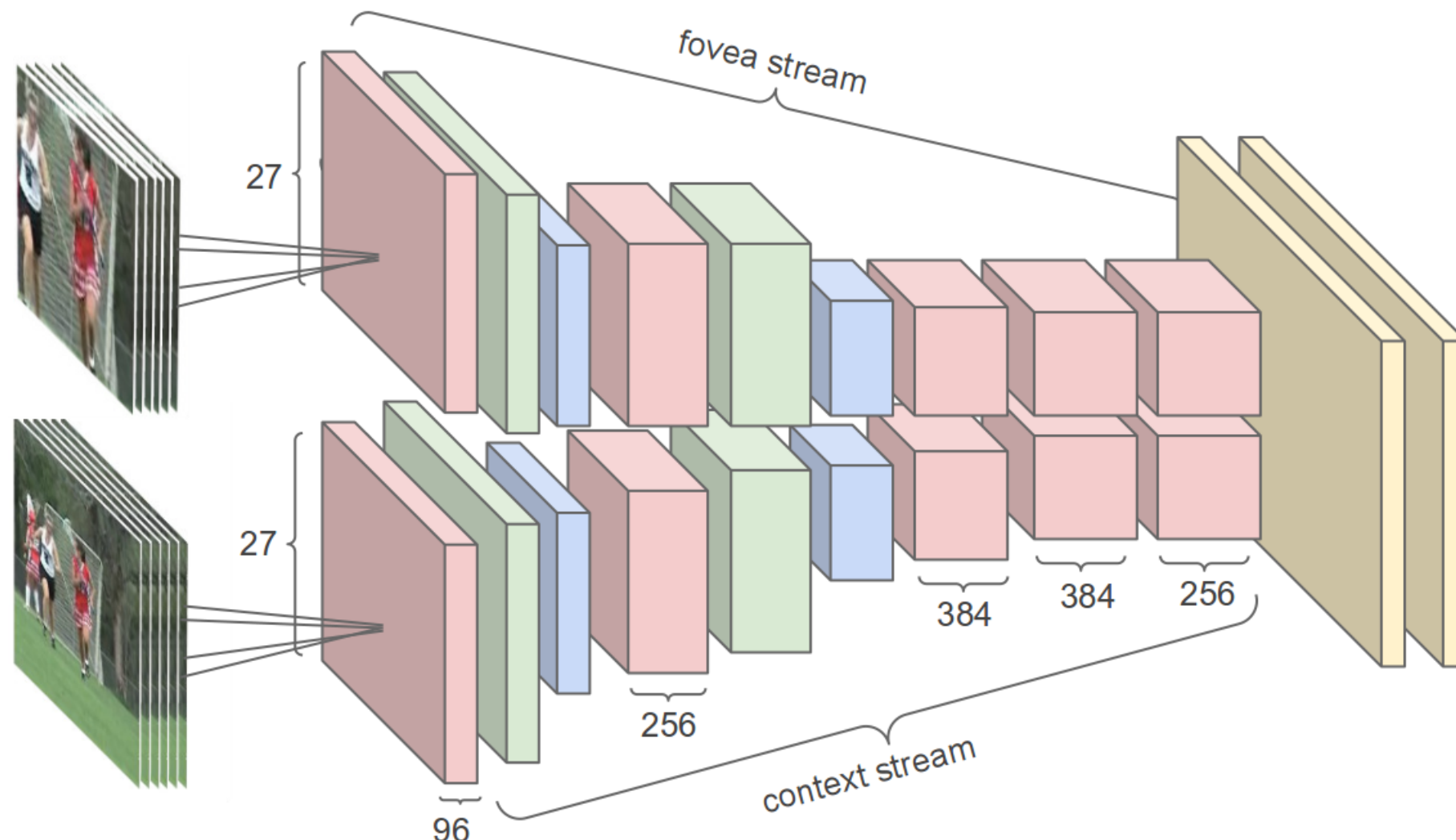


More architectures ...

- More details? Interaction? Time?
- More “complex” input data

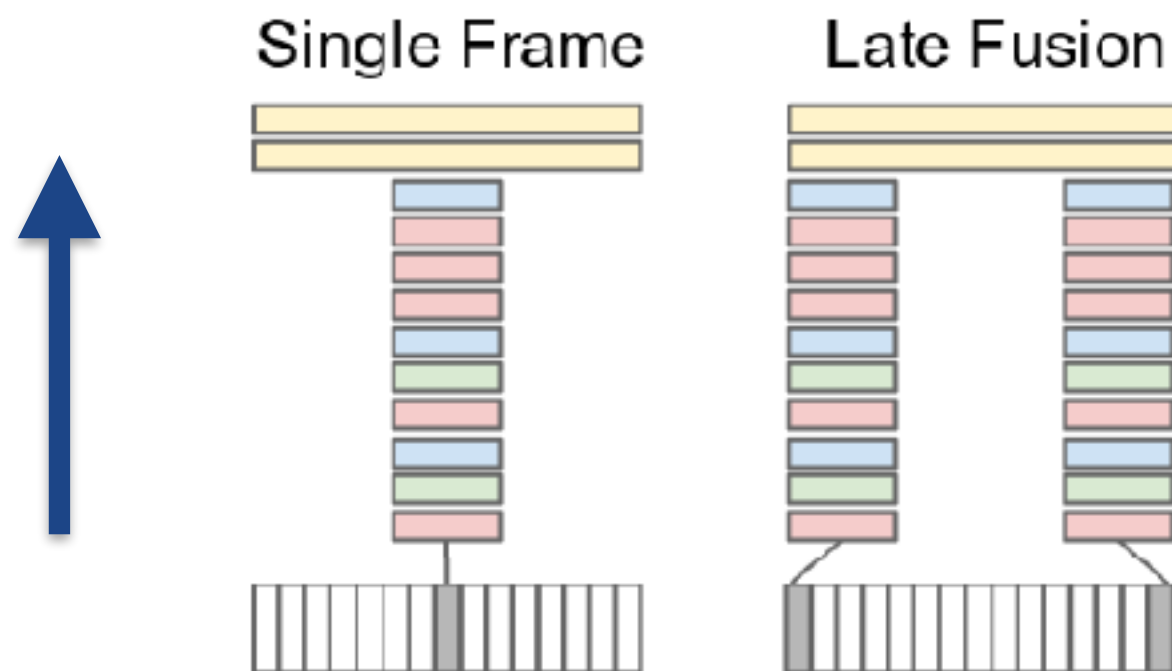
Deep Learning & Video

- Fuse multiple frame info
- high-resolution center crop + low-resolution full image (reduced input size, more efficient training)



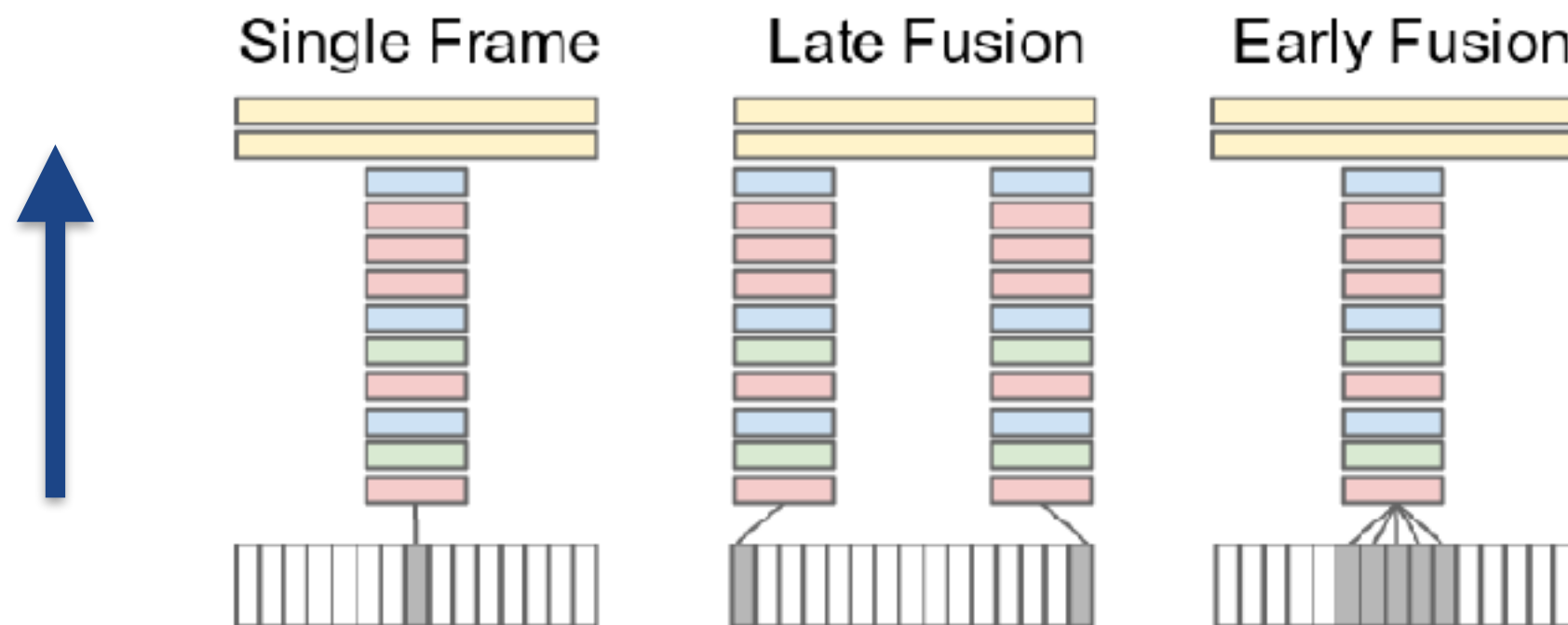
Deep Learning & Video

- Fuse multiple frame info. Many strategies



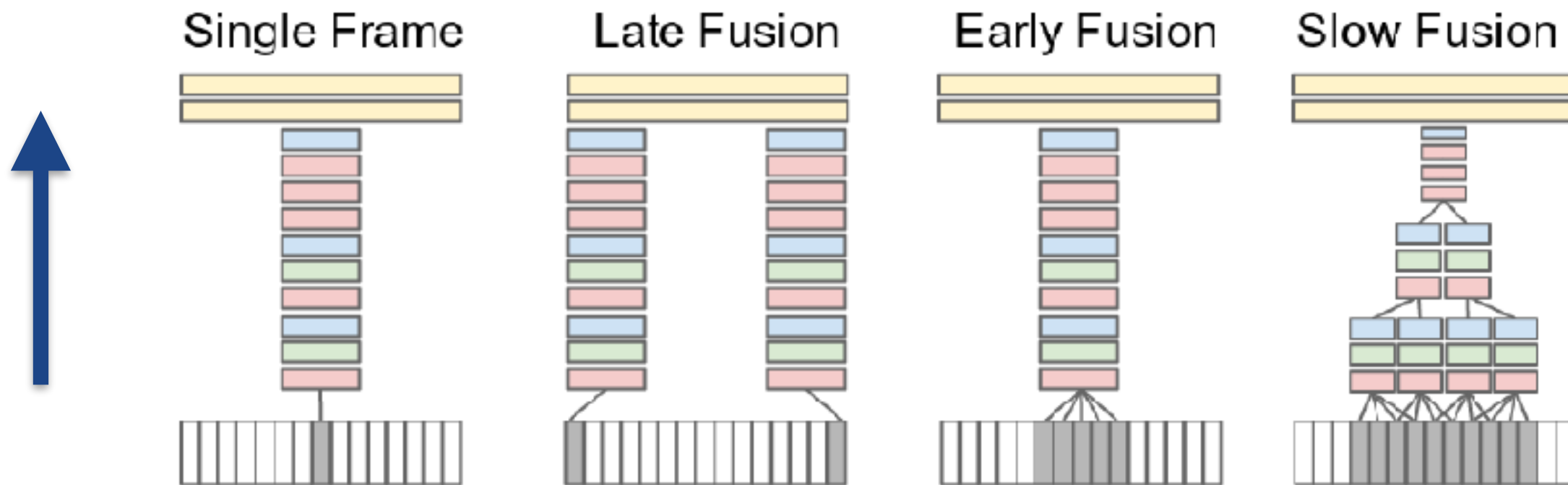
Deep Learning & Video

- Fuse multiple frame info. Many strategies



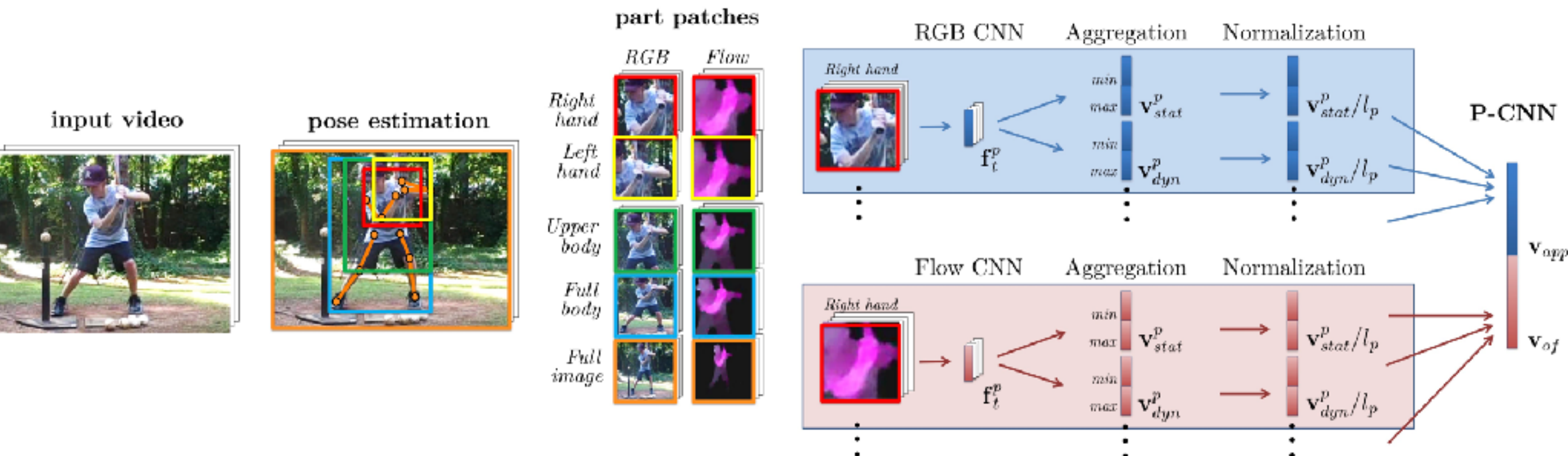
Deep Learning & Video

- Fuse multiple frame info. Many strategies



Deep Learning & Multi-Modal

- Multi-modal input from images
 - Deep learning features: Pose-based CNN
 - Combine **parts**, **pose** and **flow** with CNNs (aggregates motion and appearance information)



Deep Learning & Video

Do you see any problem with this way of treating sequential/video data?

Deep Learning & Video

Do you see any problem with this way of treating sequential/video data?

Later in the course: RNN, TCN, Transformers, ...

(More modern-adequate architectures to deal with sequential and multi-modal data)

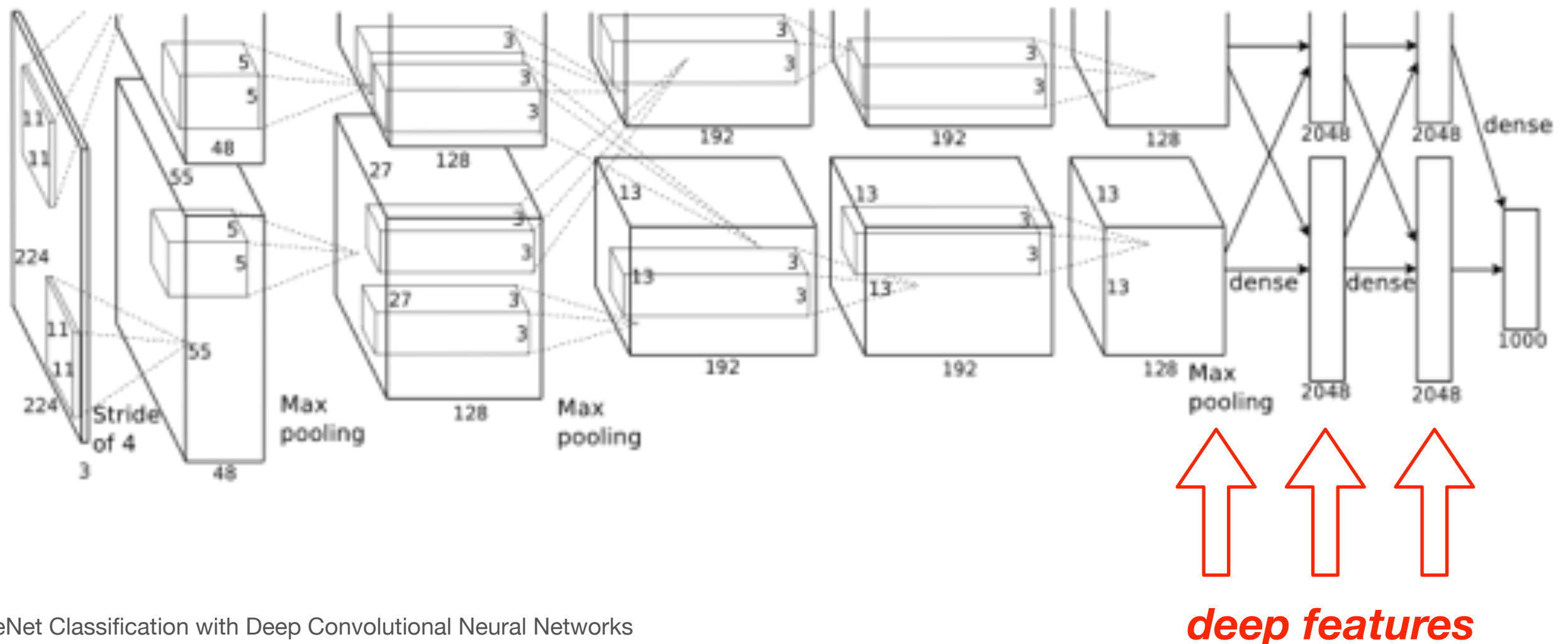
Summarizing: CNNs & Transfer Learning

- CNNs are able to generalize well!

- great **features**

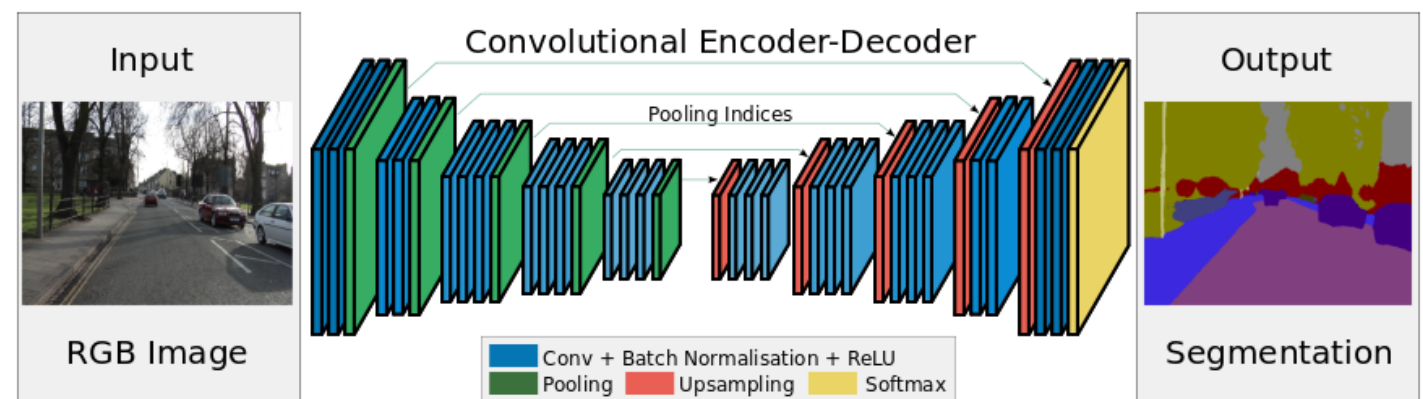
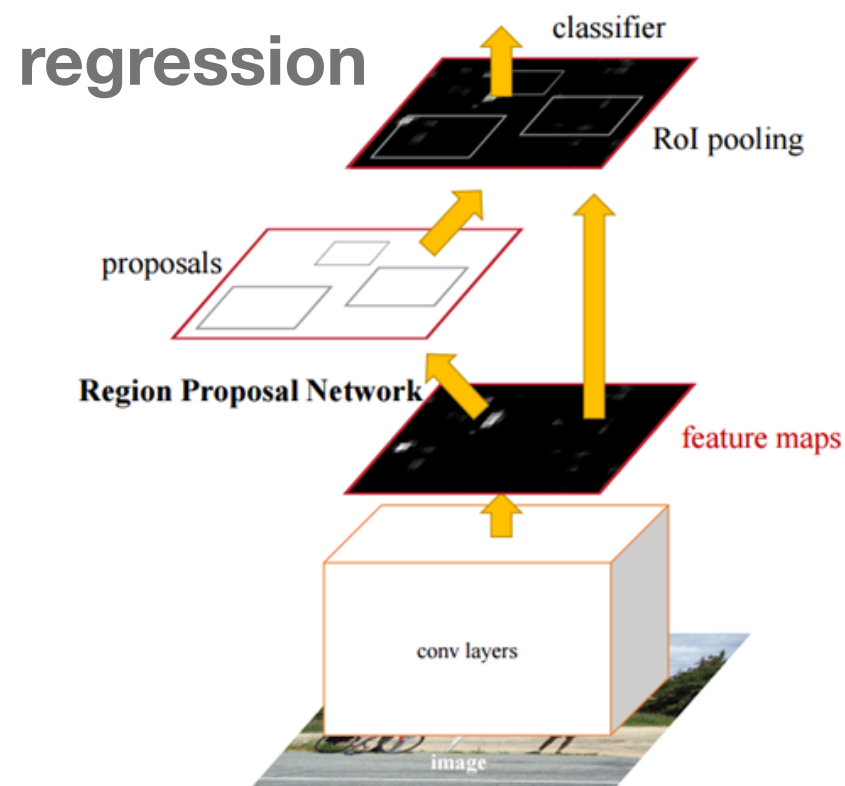
- **fine-tuning**

Lab 3
you'll practice
some of this

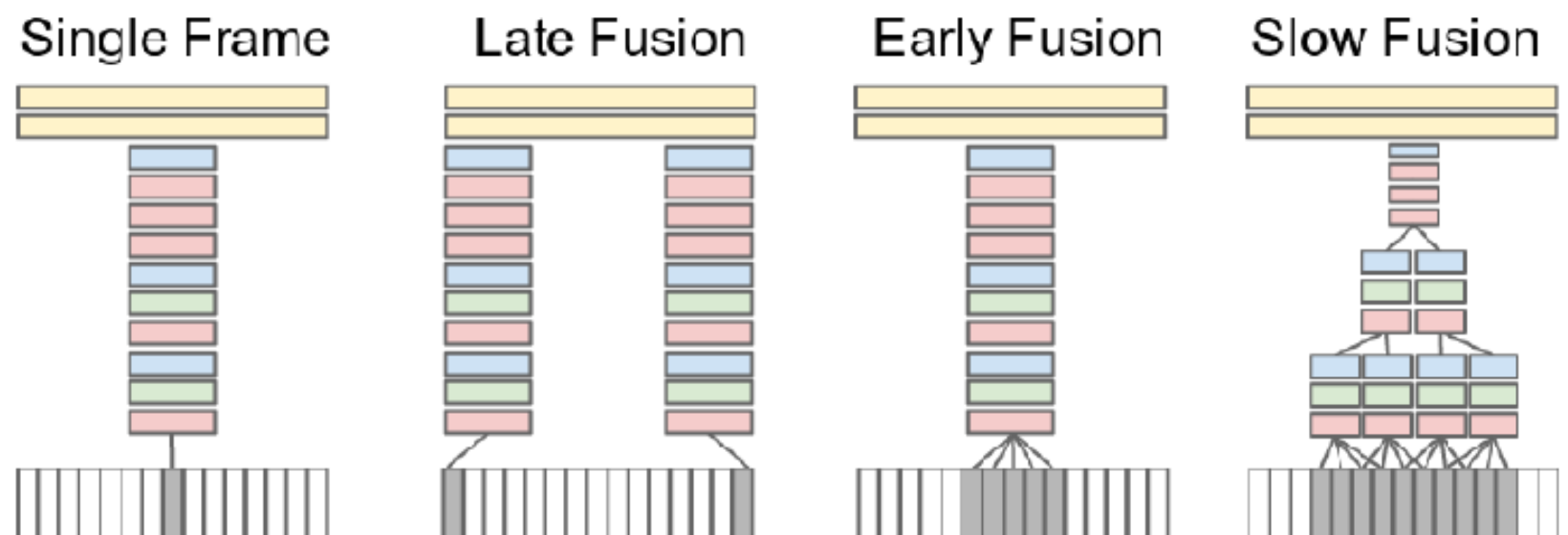


Summarizing - More architectures ...

- CNN - Not only image-classification



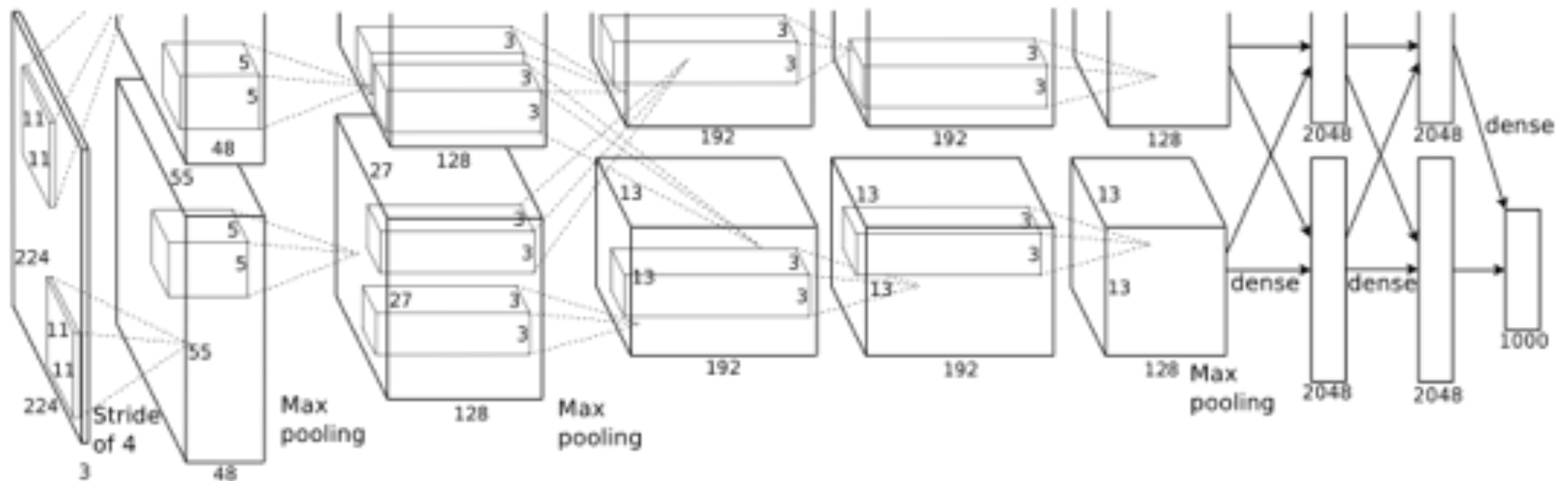
**fusion of multiple
“sources”**



Examples to understand CNNs

- **Params? feature map size?**

- How many params does the 2nd conv. layer have? and the 5th?



- Input of 240x240x3; Conv1 (48 kernels, 3x3) - Pooling (stride 2) - Conv2 (48 kernels, 3x3) —> size of feature map after Conv2?

Demos to understand CNNs

- karpathy : [DEMO online, CNN](#)
- [Playground tensorflow](#)
- visualisation (places CNN)

<http://people.csail.mit.edu/torralba/research/drawCNN/drawNet.html>

Visualizing and Understanding Convolutional Neural Networks
Matthew Zeiler and Rob Fergus. **2013**

Later ...

- More advanced models
- ... and different supervision strategies
 - DRL
 - Unsupervised
 - Recurrent architectures

Bibliography - Resources for some of the materials today

- Stanford classes on deep learning for Computer Vision (<http://cs231n.stanford.edu>) and Deep Learning (<https://cs230.stanford.edu/>)
- Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016. <http://www.deeplearningbook.org>
- Deep Learning Summer School Montreal: <https://mila.quebec/en/cours/deep-learning-summer-school-2017/>